REPUBLIC OF THE PHILIPPINES

CEBU TECHNOLOGICAL UNIVERSITY

Main Campus — Cebu City

College of Information Technology and Computer Science

CITE 201 — Programming 1

MIDTERM EXAMINATION

Second Semester, School Year 2025–2026

SCORE	/ 100
GRADE	
CHECKED BY	

Name: _____ Course & Year: _____ Date: _____

Section: _____ Instructor: _____

GENERAL INSTRUCTIONS:

Read all questions carefully before answering. Write your answers legibly. Erasures are not allowed — use correction fluid only. Strictly NO CHEATING. Any form of dishonesty will result in a grade of ZERO (0) for the entire exam. Total: 100 points.

Test I: Multiple Choice (20 pts)

Directions: **Circle** the letter of the best answer. Each item is worth **2 points**.

1. Which of the following is NOT a valid data type in C#? *(2 pts)*

A. int
B. float
C. character
D. bool

2. What keyword is used to declare a constant in C#? *(2 pts)*

A. static
B. final
C. const
D. readonly

3. What is the output of the following code? Console.WriteLine(10 % 3); *(2 pts)*

A. 3
B. 1
C. 0
D. 3.33

4. Which control structure is used to execute a block of code a fixed number of times? *(2 pts)*

A. while loop
B. do-while loop
C. for loop
D. if-else

5. What does OOP stand for? *(2 pts)*

A. Object-Oriented Programming
B. Object-Oriented Processing

C. Operational Object Protocol
D. Ordered Object Pattern

6. Which access modifier makes a class member accessible only within the same class? *(2 pts)*

A. public
B. protected
C. internal
D. private

7. In C#, which symbol is used to access a member of an object? *(2 pts)*

A. ->
B. ::
C. .
D. #

8. What is the default value of an int variable in C#? *(2 pts)*

A. null
B. undefined
C. 0
D. -1

9. Which of the following is the correct way to declare a single-dimensional array in C#? *(2 pts)*

A. int array[];
B. int[] array;
C. array int[];
D. int array[10];

10. What is the purpose of the "return" statement inside a method? *(2 pts)*

A. It ends the program
B. It exits the loop
C. It sends a value back to the caller
D. It declares a variable

Test II: True or False (10 pts)

Directions: Write **TRUE** if the statement is correct and **FALSE** if it is incorrect on the blank before each number. Each item is worth **1 point**.

_____ 1. A string in C# is a reference type.
_____ 2. The "void" return type means a method returns zero.
_____ 3. An array index in C# starts at 1.
_____ 4. Encapsulation is one of the four pillars of OOP.
_____ 5. The "break" statement can be used to exit a loop prematurely.
_____ 6. Inheritance allows a child class to use the properties and methods of a parent class.
_____ 7. A constructor must have the same name as the class.
_____ 8. The "==" operator compares values while ".Equals()" compares references for strings.
_____ 9. A "for" loop always requires an initialization, condition, and increment.
_____ 10. In C#, you can have multiple catch blocks for a single try block.

Test III: Fill in the Blanks (10 pts)

Directions: Fill in each blank with the correct word or phrase to complete the statement. Each correct answer is worth **1 point**. Spelling must be exact.

1. The process of hiding implementation details and exposing only necessary parts is called _____.

2. A _____ is a blueprint for creating objects in OOP.

3. The C# keyword _____ is used to create a new instance of a class.

4. An if-else-if chain can be replaced with a _____ statement for cleaner code.

5. The _____ loop checks its condition AFTER executing its body at least once.

6. In C#, the _____ keyword is used to define a method that can be overridden in a subclass.

7. A method that has the same name as the class and no return type is called a _____.

8. The act of a subclass providing its own implementation of an inherited method is called _____.

9. The _____ property of an array returns the total number of elements it contains.

10. An exception that is not handled will cause the program to _____ abnormally.

Test IV: Matching Type (15 pts)

Directions: Match Column A with the correct term/description in Column B. Write only the letter of your answer on the space provided. Each correct match is worth **1 point**. Each letter in Column B may be used only ONCE.

No.	Column A	Ans.	No.	Column B
1	Defines the blueprint of an object	_____	A	abstract class
2	Prevents inheritance of a class	_____	B	enum
3	The ability of objects to take many forms	_____	C	char
4	A method without a body (no implementation)	_____	D	class
5	Represents a single character value	_____	E	abstract method
6	Used to handle runtime errors	_____	F	sealed
7	Inheriting from more than one class indirectly	_____	G	polymorphism
8	A named constant block of statements that returns a value	_____	H	try-catch
9	The operator used to check if an object is null-safe in C#	_____	I	method (function)

10	Converts one data type to another explicitly	_____	J	multilevel inheritance
11	A variable that belongs to the class, not instances	_____	K	static
12	A class that cannot be instantiated directly	_____	L	casting
13	Calling the parent constructor from a child class	_____	M	?.
14	An interface member must be _____ in its implementing class	_____	N	base()
15	Used to define a group of named integer constants	_____	O	implemented (overridden)

Test V: Hand Tracing / Code Analysis (20 pts)

Directions: Trace the given code and write the **exact output** on the space provided. Show your trace work for partial credit. Each item is worth **4 points**.

Item 1.

```
int x = 5, y = 3;
int result = x * y + (x - y) * 2;
Console.WriteLine("Result: " + result);
for (int i = 1; i <= 3; i++) {
    Console.WriteLine(i + " x " + x + " = " + (i * x));
}
```

Output: _____

_____

_____

Item 2.

```
string[] fruits = {"Mango", "Banana", "Pineapple"};
foreach (string fruit in fruits) {
    if (fruit.Length > 5)
        Console.WriteLine(fruit + " has more than 5 letters.");
    else
        Console.WriteLine(fruit + " is short.");
}
```

Output: _____

_____

_____

Item 3.

```
int n = 10;
int sum = 0;
while (n > 0) {
    sum += n;
    n -= 3;
}
```

```
Console.WriteLine("n = " + n + ", sum = " + sum);
```

Output: _____

Item 4.

```
static int Factorial(int n) {
    if (n == 0 || n == 1) return 1;
    return n * Factorial(n - 1);
}
Console.WriteLine(Factorial(4));
```

Output: _____

Item 5.

```
class Animal {
    public virtual void Sound() { Console.WriteLine("..."); }
}
class Dog : Animal {
    public override void Sound() { Console.WriteLine("Woof!"); }
}
Animal a = new Dog();
a.Sound();
```

Output: _____

Test VI: Programming / Problem Solving (15 pts)

Directions: Write a complete and working C# console program for each problem. Follow proper naming conventions, indentation, and syntax. Partial credit may be given for correct logic even with minor syntax errors. Each item is worth **5 points**.

Problem 1. (5 pts)

Write a C# program that asks the user to input 5 integers and displays their sum, average, highest value, and lowest value.

Answer / Solution:

Problem 2. (5 pts)

Create a class called "Student" with the following attributes: name (string), studentNumber (string), and GPA (double). Include a constructor, appropriate getters and setters, and a method called

DisplayInfo() that prints all student information in a formatted manner. Instantiate the class in the Main method and call DisplayInfo().

Answer / Solution:

Problem 3. (5 pts)

Write a C# program that reads 10 integers from the user, stores them in an array, and uses a method called BubbleSort() to sort the array in ascending order. Display the unsorted and sorted arrays.

Answer / Solution:

Test VII: Essay / Short Answer (10 pts)

Directions: Answer the following questions in **3 to 5 sentences** each. Be concise, clear, and use proper technical terminology. Each item is worth **5 points**.

Question 1. (5 pts)

Explain the concept of Object-Oriented Programming (OOP) and briefly describe each of its four fundamental principles: Encapsulation, Abstraction, Inheritance, and Polymorphism. Give a real-world analogy for any one of the four principles.

Question 2. (5 pts)

Differentiate between a "for" loop, a "while" loop, and a "do-while" loop in C#. Describe a specific scenario or use case where each type of loop is the most appropriate choice, and explain why.

— END OF EXAMINATION — Good luck! —

"The secret of getting ahead is getting started." — Mark Twain

ANSWER KEY

CITE 201 — Programming 1 | Midterm Examination | S.Y. 2025–2026

FOR INSTRUCTOR USE ONLY. Do not distribute to students.

Test I: Multiple Choice (20 pts — 2 pts each)

1. C — character *— not a valid C# keyword; use 'char'*
2. C — const
3. B — 1 *— 10 mod 3 = 1*
4. C — for loop
5. A — Object-Oriented Programming
6. D — private
7. C — . (dot operator)
8. C — 0
9. B — int[] array;
10. C — It sends a value back to the caller

Test II: True or False (10 pts — 1 pt each)

1. TRUE *— string is a reference type*
2. FALSE *— void means no return value*
3. FALSE *— index starts at 0*
4. TRUE
5. TRUE
6. TRUE
7. TRUE
8. FALSE *— for strings, == compares values in C#*
9. FALSE *— parts are optional*
10. TRUE

Test III: Fill in the Blanks (10 pts — 1 pt each)

1. Encapsulation
2. class
3. new
4. switch
5. do-while
6. virtual
7. constructor
8. method overriding
9. Length
10. terminate / crash

Test IV: Matching Type (15 pts — 1 pt each)

1. D — class
2. F — sealed
3. G — polymorphism
4. E — abstract method
5. C — char
6. H — try-catch
7. J — multilevel inheritance
8. I — method (function)
9. M — ?.
10. L — casting
11. K — static
12. A — abstract class
13. N — base()
14. O — implemented (overridden)
15. B — enum

Test V: Hand Tracing / Code Analysis (20 pts — 4 pts each)

Item 1:

Result: 21
1 x 5 = 5
2 x 5 = 10
3 x 5 = 15

Item 2:

Mango has more than 5 letters.
Banana has more than 5 letters.
Pineapple has more than 5 letters.

Item 3:
n = -2, sum = 22
(Trace: 10+7+4+1 = 22; n ends at -2)

Item 4:
24
(4! = 4 x 3 x 2 x 1 = 24)

Item 5:
Woof!
(Runtime polymorphism — overridden method is called)

Test VI: Programming / Problem Solving (15 pts — 5 pts each)

Grading Rubric per Problem:

Points	Criteria	Score
2 pts	Correct logic and algorithm (even with minor syntax errors)	___ / 2
1 pt	Correct class/method structure and proper naming conventions	___ / 1
1 pt	Correct input/output handling	___ / 1
1 pt	Proper indentation and code readability	___ / 1

Test VII: Essay / Short Answer (10 pts — 5 pts each)

Points	Criteria	Score
2 pts	Accuracy and completeness of technical content	___ / 2
1 pt	Clarity and organization of explanation	___ / 1
1 pt	Correct use of technical terms and vocabulary	___ / 1
1 pt	Quality of real-world analogy or example given	___ / 1

Section	Max Points	Student Score	Remarks
I — Multiple Choice	20		
II — True or False	10		
III — Fill in the Blanks	10		
IV — Matching Type	15		
V — Hand Tracing	20		
VI — Programming	15		
VII — Essay	10		
TOTAL	**100**	_____	

— END OF ANSWER KEY —